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Basics of stereoscopic video, new compression results with MPEG-2 and a proposal for MPEG-4

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Abstract

Many current as well as emerging applications in areas of entertainment, remote operations, manufacturing industry and medicine can benefit from the depth perception offered by stereoscopic video systems which employ two views of a scene imaged under the constraints imposed by the human visual system. Among the many challenges to be overcome for practical realization and widespread use of 3D/stereoscopic systems are good systems for 3D video capture, display and efficient techniques for digital compression of enormous amounts of data while maintaining compatibility with normal video decoding and display systems.

After a brief introduction to the basics of 3D/stereo including issues of depth perception, imaging and display, we present a brief overview of portions of the MPEG-2 video standard that are relevant to our discussion on compression of stereoscopic video. Next, we outline the various approaches for compression of stereoscopic video and then focus on compatible stereoscopic video coding using MPEG-2 Temporal scalability concepts. Compatible coding employing two different types of prediction structures become potentially possible, disparity compensated prediction and combined disparity and motion compensated predictions. To further improve coding performance and display quality, gain and offset preprocessing for reducing mismatch between the two views forming stereoscopic video is considered. We then introduce the various considerations in coding of stereoscopic video at lower bit-rates for the ongoing MPEG-4 standard. A method is proposed that builds on the proven framework of MPEG-2 like coding but introduces additional coding flexibilities to achieve reasonable performance at lower bit-rates for MPEG-4.

Next, results of experiments are presented for a variety of combinations of MPEG-2 based coding methods for the left and the right views while employing TV resolution video for a number of sequences and for various bit-rates. The combined disparity and motion-compensated prediction is found to offer the best performance among combinations tested. These results indicate that compression of both views of stereoscopic video of normal TV resolution appears feasible with good quality in a total of 6–8 Mbit/s. Further, results are presented at much lower bit-rates based on the coding method proposed for MPEG-4 on two long test sequences. We then discuss multi-viewpoint video applications, the ongoing efforts towards a multi-viewpoint profile in MPEG-2 and expected direction of multi-viewpoint video coding in MPEG-4. © 1997 Elsevier Science B.V.

Keywords: Stereoscopic video; 3D video; Video compression; Temporal scalability; Stereoscopic video coding; MPEG-4 standard; MPEG-2 standard; Multi-viewpoint video

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1. Introduction

Although the human visual system can naturally capture/interpret the three-dimensional real world, most imaging and display systems limit our abilities by presenting only a two-dimensional (2D) mapping of the three-dimensional (3D) world. Many current as well as emerging applications can significantly benefit from the high degree of realism provided by 3D scenes with depth. In fact, human ability to perceive realism in a scene is directly related to our ability to perceive depth accurately in a scene. A real 3D scene can be reconstructed from many 2D views of a scene imaged from different perspectives. Such a representation would not only allow the perception of depth in a scene but look-around capability as well. However, due to practical display constraints, only a less complex solution employing two views of a scene is easily realizable; it can still impart the sensation of depth. These two views are required to be imaged under specific constraints and are intended, one for each eye of a human visual system, so that our brain can generate the depth information necessary to perceive realism. The two views when captured together constitute stereoscopic video, where each captured view is similar to normal video except for the fact that they are related under constraints imposed by binocular vision.

Stereoscopic video has potential applications in education, training, 3D movies/entertainment, medical surgery, videoconferencing, virtual travel/shopping, multimedia presentations, video games, immersive virtual reality experiences and others [5, 18]. Although, the potential applications of 3D/stereoscopic video are many, there are several challenges to be overcome before its potential can be truly harnessed and its use becomes wide spread. For example, currently, most practical means of displaying stereo require viewers to use specialized viewing glasses. Although some displays not requiring specialized viewing glasses (autostereoscopic systems) are available, they impose other restrictions, e.g., viewing zones and discreteness of views. Typically, the two views of a scene in stereoscopic video, where one is referred to as the left view and other is referred to as the right view, are intended for presentation to the left eye and the right eye of a human visual system in either time-sequential (with active synchronized shuttered glasses) or time-simultaneous (with passive polarizing glasses or other optical barriers). Besides the display issue, the other main issue is that of efficient digital compression of 3D/stereoscopic video so that a large volume of generated data can be easily manipulated, stored or transmitted as needed. Towards that end, interworking with existing or emerging coding schemes as well as existing displays for normal video may be a necessity in many applications.

After many fits and starts, the era of digital TV is finally here with the recent introduction of digital direct broadcast satellite as well as digital Cable TV. The video compression method used in both these services is based on the second phase standard of the ISO Moving picture Experts Group (MPEG-2). In terms of bandwidth needed for compression, MPEG-2 encoding requires 3–8 Mbit/s for consumer-grade quality and 9–15 Mbit/s for higher quality. Further in the US, for digital terrestrial broadcast of HDTV, the FCC advisory committee on advanced TV (ACATS) has recommended MPEG-2 video coding. With the current digital TV and HDTV at the brink of introduction, the next logical step is TV offering a sense of depth, which is referred to as 3DTV [5, 18]. Based on experience from temporal scalability coding of MPEG-2, 3DTV is expected to take bandwidth of no more than one-and-half times the bandwidth of one video signal.

In this paper, we examine the two main issues in the practical realization of 3DTV; they are, a practical display, and, efficient compression of stereoscopic video. We will not discuss the issue of 3D displays in detail but only cover the basics necessary to understand the operation of stereoscopic 3D display. In addition, we will review the terminology used in stereoscopic imaging and display. The issue of efficient compression of stereoscopic video in a manner compatible with existing TV displays is considered. Of particular relevance to our discussion is MPEG-2 video coding [21, 24], which, in addition to applications of digital direct broadcast satellite TV, Cable TV, terrestrial HDTV mentioned earlier is also useful for digital VCR, video-on-demand, multipoint video and many other applications. Thus, employing MPEG-2 for

coding of stereoscopic video seems particularly meaningful due to the potential of interoperability with the rest of the future digital TV infrastructure. Furthermore, for inexpensive consumer applications such as video games, we consider stereoscopic video compression at lower resolution and bit-rates within the context of MPEG-4, the next MPEG standard which at present is ongoing.

In the next section, we discuss the basics of 3D/stereo including cues that assist in depth perception, the important issue of stereoscopic 3D displays, and the geometry of stereoscopic imaging and display systems. In Section 3, we discuss the basics of MPEG-2 video coding with special emphasis on temporal scalability. Next, Section 4 outlines the various possible approaches for stereoscopic video coding. In Section 5, we explain how the temporal scalability tool of MPEG-2 can be employed for compression of stereoscopic video in a compatible manner. Section 6 presents our preprocessing approach that can improve not only coding efficiency while maintaining compatibility with MPEG-2 video coding but also improve quality of displayed stereoscopic video. In Section 7 we introduce issues and considerations for stereoscopic coding for MPEG-4 followed by our proposed generalized approach. Section 8 presents results of experiments using MPEG-2, with and without preprocessing as well as results of our proposal for MPEG-4. Section 9 discusses multiviewpoint video, its applications and systems description, followed by currently existing multiviewpoint coding capabilities in MPEG-2 and anticipated directions for MPEG-4. Finally, Section 10, summarizes the main issues discussed in this paper.

2. 3D/stereoscopic vision, imaging and display basics

In this section, we first present an overview of stereoscopic terminology, following which we introduce the basics of depth perception, and lastly discuss the issues of 3D/stereoscopic imaging and display.

The human visual system consists of a pair of eyes, a left eye and a right eye. The back of each eye contains a sensing element called the retina that is connected via optic nerves to a region called the lateral geniculate body, and via pathways to the visual cortex in the brain. The image produced on each of the two retinas is upside down, however, significant processing takes place beyond the retina. The nerves from right retina cross over to the left side lateral geniculate body and nerves from left retina cross over to the right side lateral geniculate body, at the underside of the brain. Some information processing takes place along the way. Overall, the end result of all this processing is the appearance of a single upright image providing a sense of depth.

2.1. Visual depth cues

We learn from the world around us by experience. The outcome of that experience is a huge database of visual information that human brain accumulates. This lets us perform everyday tasks, many of which involve judging relative size of objects, shapes, distances, colors and much more. The ability to accurately perceive depth in the 3D world is an important attribute of the human visual system. There are many cues that assist us in this task. They can be classified in at least two different ways: monocular versus binocular, and physiological versus psychological.

2.1.1. Physiological cues

We now list with a brief explanation some of the various physiological cues.

- *Accommodation* is the change in the focal length of the lens in the eye caused by muscles in the eye to produce a focused image on the retina.
- *Convergence* is the rotation of eyes to align or merge the left and right eye images into a single image with perceived depth.
- *Binocular disparity* is the dissimilarity in views due to the relative location of each eye.
- *Motion parallax* is the difference in views produced by moving the scene or the viewer.
- *Chroma-stereopsis* is the difference in apparent depth due to the color of an object from refraction effects in the eyes.

2.1.2. Psychological cues

Next, we list with a brief explanation some of the various psychological cues.

- *Image size* is useful but not sufficient to determine size or depth of objects.
- *Linear perspective* is the decrease in the apparent size of an object with increasing distance.
- *Aerial perspective* refers to the hazy and bluish appearance of distant objects.
- *Shading* suggests that objects farther from the source of light are darker.
- *Shadowing* of an object on others provides clues about position and size.
- *Occlusion* of objects provides a clue about their relative location.
- *Brightness* of an object suggests that it is closer than dimmer objects.
- *Texture gradient* provides clues regarding distance and relative location.

2.1.3. Other cues

Besides physiological and psychological cues there are other cues that assist us. *Experience* lets us estimate relative size, shape, distance and color of objects.

Overall, the many cues listed are generally additive and all contribute to the process of 3D vision. However, some cues are more powerful than others and this may lead to contradictory depth information. A good 3D imaging system will minimize contradictory depth cues. In general, binocular disparity is a sufficient but not necessary cue needed to perceive depth in a scene. That it is a sufficient condition is readily demonstrated by random dot and 'magic eye' stereograms in which the only depth cue is binocular disparity. That it is not necessary is demonstrated by the fact that the world does not go flat when one closes one eye.

2.1.4. More on binocular cues

As mentioned earlier, an alternative classification of visual depth cues is possible based on whether they are binocular or monocular. Due to the special relevance of binocular cues to our discussion, we provide further explanation.

Accommodation, as discussed earlier, consists of the adjustment of the focal length of the lens in the human eye by the surrounding muscles; the lens, in effect, changes shape becoming thicker or flatter as needed to focus on a distant object. In contrast, the lens in a camera does not change its shape, but moves relative to the image plane to produce a focused image.

Convergence consists of aligning the images of the left and right eye by rotating each eyeball and thus adjusting the optic axis of each eye to get the two axes to meet at a fixation point. The angle between the optic axis of the two eyes is called the *convergence angle*.

In normal viewing, accommodation and convergence usually work together, for example, if an object suddenly moves from a distance towards us, our eyes quickly change the focus and the orientation in an attempt to track the movement of the object. In many 3D imaging systems, the eyes must focus on a screen surface while the convergence is determined by the apparent position of the object in front or in back of the screen. These contradictory depth cues can prevent the viewer from merging the left and right eye images or may produce eye strain during prolonged viewing.

2.2. Techniques for 3D viewing

We are now ready to address the first issue in realization of practical 3DTV, the issue of displaying in 3D. In normal human vision, binocular disparity is the main cue responsible for the perception of depth in scenes. Each eye of a pair of eyes of a human visual system sees a slightly different view of the world around us and the human brain fuses these two slightly different views resulting in a single view with depth. For viewing captured scenes with the sense of depth some type of display arrangement capable of displaying independent views to each eye is required.

There are various approaches for classifying 3D displays. According to one approach, 3D displays can be classified into three classes: *volumetric* 3D, *stereoscopic* 3D and *perspective* 3D. *Volumetric* 3D displays recreate a pattern of light rays in actual 3D space. Examples of this type of display are real scenes, holograms, rotating screen and oscillating mirror displays. *Volumetric* displays have the

advantage that they can be viewed by multiple viewers from any angle (or at least over a wide range of angles) without special glasses. Their disadvantage is that they are for the most part restricted to displaying computer-generated graphics. Most are restricted to displaying wire frame renderings and cannot deal with hidden surfaces. At present there is no camera which can acquire live video for *volumetric* 3D display. The commercial availability of *volumetric* 3D displays is also limited. *Stereoscopic* 3D displays recreate pairs of images, one intended for the left eye and the other intended for the right eye. Stereo pairs can be acquired in full color by conventional video cameras and displayed on modified conventional displays. They provide only horizontal disparity unless head tracking is provided for live or computer-generated graphics. *Stereoscopic* 3D systems are readily available commercially at a reasonable cost. *Perspective* 3D displays present the same images to both eyes and make use of depth cues other than binocular disparity. In most computer graphics displays, 3D graphics only refers to *perspective* 3D. *Perspective* 3D displays can be quite effective in producing a sensation of depth especially with large high-resolution displays which cover the entire visual field of the viewer. In these large screen displays, the apparent sensation of depth arises from the lack of conflicting depth cues such as screen texture and surround. However, the size and resolution of the display required to obtain a convincing 3D effect from a *perspective* 3D system is prohibitive except in a commercial theater setting.

Using a different approach, 3D displays can also be classified as autostereoscopic and stereoscopic. Autostereoscopic displays do not require special viewing glasses or other optical arrangement while stereoscopic displays require special viewing glasses or other optical arrangement to allow 3D viewing. From a practical standpoint, among the various displays for viewing in 3D with good quality, stereoscopic displays are the most mature and the most cost-effective thus far. Autostereoscopic displays suffer from the disadvantage that they can typically be viewed from only one or a few position making them impractical for large audiences.

2.3. Stereoscopic display

In stereoscopic viewing, two sets of images are used, one offered to the left eye and the other to the right eye of the viewer. Each of the right eye and the left eye views are normal two-dimensional views as may be achieved by imaging using two separate cameras. Furthermore, the two views are imaged under a specific constraint, such that the human brain can fuse these two slightly different views resulting in the sensation of depth in the scene being viewed. The principal characteristic of a stereoscopic display is that it requires viewers to use some sort of specialized glasses such as red-and-blue glasses or polarized glasses, or other optical arrangements such as head-mounted displays (HMD).

2.3.1. Passive separation

When light is filtered in such a manner that all of the photons have their electric field vectors aligned in the same direction, it is said to be linearly polarized. Light may also be circularly polarized by the combination of a linear polarizer and a quarter-wave retarder. For circularly polarized light the electric field vector rotates in a clockwise or counterclockwise direction. Passive 3D displays usually employ projection systems in which the left and right eye images are projected on to a screen with different polarizations or colors. When polarized light is used, the screen must have a metallic surface to preserve the incident polarization state in the reflected light.

Color separation techniques involve imaging the left and right views in non-overlapping spectral bands, usually red and blue. The viewer then wears glasses with band pass filters which only allow light to reach the appropriate eye. The major disadvantage of this technique is that it is limited to monochrome imaging, i.e. for red and blue filters the viewed image is in magenta. This simple low cost method of obtaining 3D is especially useful in aligning pairs of stereo cameras. The camera outputs can be viewed on a single color monitor to remove vertical disparity. This usual color separation technique should not be confused with the chromastereopsis effect which arises from the fact that red and blue light will not focus at the same point on

the retina and causes most individuals to perceive an apparent difference in depth of highly saturated red and blue images. Chroma-stereopsis is very viewer dependent with some observers seeing no effect. However, prismatic glasses have been developed which use the underlying physical principal of the chroma-stereopsis to permit the viewing full color 3D images.

Linear polarization techniques for 3D displays use linearly polarized light in two orthogonal directions, say, horizontal and vertical, to separate the images intended for the left and right eye. The viewer wears glasses with linear polarizing filters which block light polarized orthogonal to the filter axis. It suffers from the drawback that tilting of the head by the viewer causes misalignment between the direction of polarized light coming from screen and the corresponding orthogonal filter in the glasses. This produces cross talk between the images intended for the left and right eyes. Tilting of the head by only a few degrees can be sufficient to produce enough cross talk to destroy the 3D effect.

Circular polarization offers an alternative to the limitations of linear polarization. In circular polarization, two different polarization states, one clockwise and the other counterclockwise are used. Circular polarizing filters are placed over each eye which filters out the image intended for the opposite eye. Circular polarizing filters are a bit more expensive than linear polarizing filters and do not have as high an extinction ratio as two crossed linear polarizers. However, they have the distinct advantage that much less cross talk occurs as the viewer's head is tilted. Since most 3D displays only provide static horizontal binocular disparity, most viewers will lose the ability to merge the right and left eye images when the head is tilted more than about 20° even with no cross talk.

2.3.2. Active separation

This technique is used with active liquid crystal (LC) shutter glasses which are synchronized with a display which may be a TV monitor or a projection system. The LC glasses act as an alternating electronic shutter such that when the shutter closes over one eye preventing any light from reaching it, it opens on the other eye, allowing light to fully pass through it. If the left and the right view frames are alternatingly multiplexed in time, the glasses will separate the images intended for the left and right eyes. Since these glasses contain the electronics needed for shuttering, they are called active glasses as compared to passive glasses which contain no electronics. It should be noted however that due to the need for temporal multiplexing of the frames of two views, the display must run at twice the speed in order for each eye to be able to view video at normal temporal resolution without flicker. Active techniques have the advantage that they require only a single display or projector. Polarization-preserving screens are not needed and no cross talk is introduced by head tilt. A further disadvantage of active systems is the cost of the glasses and their maintenance which may require batteries. The glasses must also obtain a synchronization signal from the display either through a wired or wireless link.

2.4. Stereoscopic imaging and display geometry

Now, we briefly review some of the terms typically used in stereoscopic imaging and display. Fig. 1(a) shows the typical setup employed in stereoscopic imaging, while Fig. 1(b) shows an example of the setup used for stereoscopic display.

- *Convergence point* is the point where the optic axes of the left and the right eyes or left and right cameras meet. This point is also called the object point.
- *Convergence distance* is the distance between midpoint of line passing through the camera lenses and the convergence point. This is also called the pick-up distance.
- *Interocular distance* is the distance between the right and the left eye.
- *Camera separation* is the distance between the centers of the camera lenses. This is also called the baseline distance.
- *Viewing distance* is the distance between the viewer and the screen.
- *CCD sensor width* is the width of the charge-coupled device used as the image sensor in the video camera.

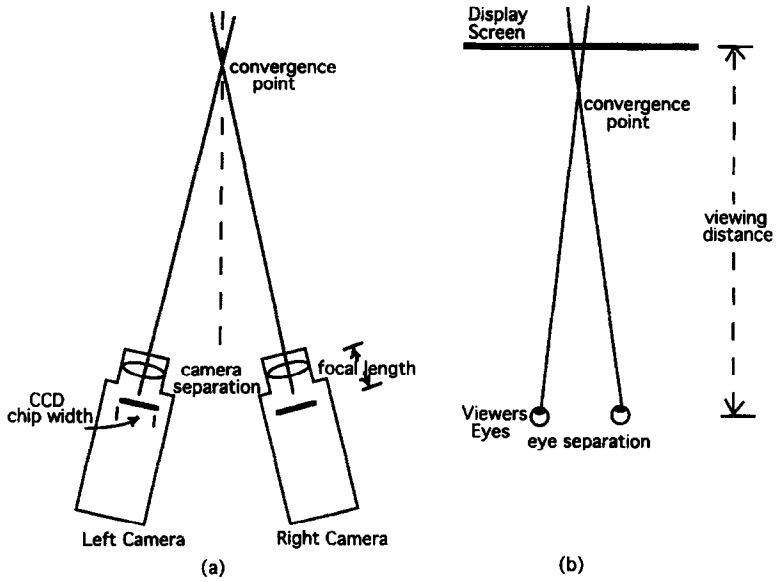

Fig. 1. Camera and display geometries.

3. MPEG-2 video coding basics

Now that we have an understanding of basic terminology and issues in 3D/stereoscopic imaging and display, we need to briefly review portions of MPEG-2 video coding due to their direct relevance to techniques we use for compression of stereoscopic video.

The MPEG-2 video standard [21, 24, 26] is based on the proven waveform-coding-based framework employing interframe block motion-compensated-DCT coding. It is especially optimized for interlaced video. The standard specifies a bit-stream syntax and decoding semantics [24] only, allowing considerable innovation in optimization of encoding [10] to improve coding quality. As is well recognized, the MPEG-2 video standard, not only offers a good solution to the problem of providing high-quality video for broadcast applications at reasonable bit-rates, but also includes a useful set of tools allowing scalable coding of video for applications in ATM, multipoint video, HDTV, multimedia retrieval and others. Earlier, [13], it was envisaged that some of the scalability tools of MPEG-2, such as spatial scalability and temporal scalability [19], were expected to be particularly relevant to the coding of stereoscopic video, although at that time, confirmation by experiments was incomplete. We now present a brief overview of a few issues in MPEG-2 compression relevant to our discussion on stereoscopic coding.

3.1. Nonscalable coding

By nonscalable video coding we mean normal coding of video as a single layer. MPEG-2 supports nonscalable coding of video with high efficiency as well as scalable coding of video. MPEG-2 video coding is designed to be a superset of MPEG-1 video coding and thus supports nonscalable coding of non-interlaced video but is specially optimized for coding of interlaced video with high quality. We now briefly review the basic terminology used in MPEG-2 video coding and its meaning.

3.1.1. Frame and field pictures

If an interlaced sequence is coded as single frames each consisting of two fields, it is said to be composed of frame-pictures. If the two fields of each frame are coded separately (but still occur in pairs) the sequence is said to be composed of field-pictures. In MPEG-2, it is also possible to combine frame-pictures and field-pictures during coding of a sequence.

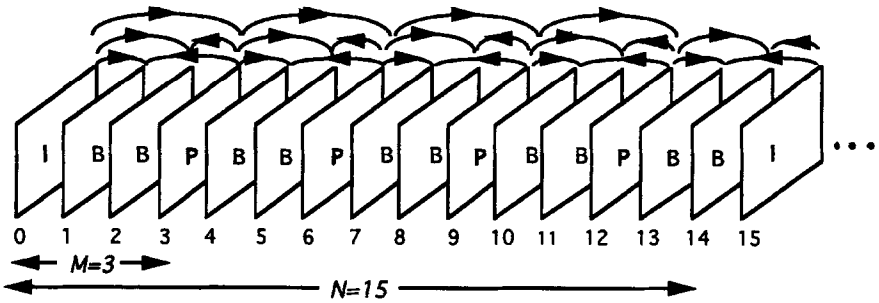

Fig. 2. Group of pictures with prediction distance, $M = 3$, and intra distance, $N = 15$.

3.1.2. Picture types and group of pictures

In MPEG-2 video coding, three types of pictures are allowed and are referred to as intra (I-), predictive (P-) and bidirectional predictive (B-) coded pictures. I-pictures are coded by themselves without referring to any past decoded pictures and serve as entry points in the case of random access. P-pictures are causal and are coded using motion-compensated prediction with respect to past decoded I- or P-pictures, whereas B-pictures are noncausal and are coded using motion-compensated prediction with respect to a past and a future I- or P-picture. A group of pictures is an optional organizational layer, and when it is present a group of pictures header precedes an I-picture. The prediction distance between consecutive reference pictures, which can be either a pair of P-pictures, an I-picture and a P-picture, or a P-picture followed by an I-picture, is referred to as 'M', and the distance between I-pictures, is referred to as 'N'. Thus, 'M' is one more than number of B-pictures between consecutive reference pictures, for example, $M = 3$ implies two B-pictures between consecutive reference pictures. Since neither the prediction distance nor the distance between I-pictures needs to be constant for all pictures in a sequence, no global information about 'M' or 'N' is carried in a bitstream but can be locally derived when decoding each picture. An example of a group-of-pictures showing prediction distane, $M = 3$, and intra distance, $N = 15$, is shown in Fig. 2.

3.1.3. Motion-compensated prediction

As is well known, motion-compensated prediction is employed to exploit temporal redundancies in interframe video coding. Motion-compensated prediction modes in MPEG-2 are built on the motion-compensated prediction modes of MPEG-1 and are further optimized to yield high prediction efficiency for interlaced video. Each 16×16 luminance block and its two associated 8×8 chrominance blocks are referred to as a macroblock. A motion vector of up to half pixel accuracy is used for predicting the position of each 16×16 luminance block. This vector is scaled for predicting the location of the corresponding chrominance blocks.

In MPEG-2, for coding of interlaced video as frame-pictures, three basic motion prediction types are allowed. The first is called frame motion type and uses 1 motion vector for each 16×16 macroblock very similar to that in MPEG-1. The second is field motion type and uses 2 motion vectors one for each of the two 16×8 field blocks that form the 16×16 block. The third is the dual-prime motion type and uses a single field motion vector that is scaled and applied along with a small correction motion vector to compute prediction for each of the two 16×8 field blocks that form the 16×16 block. The dual-prime motion type is only allowed for prediction between P-pictures where there are no B-pictures between predicted and reference frames. A 2 bit code per predicted macroblock identifies the motion prediction mode used. The decision of which motion-compensated prediction mode to choose is allowed on a macroblock basis. Since the prediction mode is an encoding issue, it is not standardized. Typically, the minimum mean square prediction error is used, although a better prediction mode decision criterion is certainly possible.

3.1.4. Macroblock modes

For P-frame pictures, macroblock modes include the ones with forward-motion-compensated prediction, no motion prediction, intra as well as combinations with specified quantizer parameters. However, for B-pictures, not only various modes of P-pictures are allowed, but a reference direction is also included when appropriate, such as forward, backward and bi-directional prediction (interpolation). The details of these modes are specified in [10, 19] and will not be discussed here.

3.1.5. DCT coding

MPEG-2 video coding like MPEG-1 video coding uses DCT coding of 8×8 blocks to exploit spatial redundancies. For frame-pictures, the DCT can be applied on either 8×8 frame blocks or on 8×8 field blocks resulting in frame-DCT type or field-DCT type which is identified by a 1-bit code on a macroblock basis.

3.2. Temporal scalability

Originally, temporal scalability was introduced in MPEG-2 as a migration path to high-resolution progressive HDTV from the interim solution of lower temporal resolution interlaced or progressive HDTV. We now provide a brief overview of the basics of temporal scalability [13]. Temporal scalability in MPEG-2 refers to layered coding with 2 or more layers, each with the same or different temporal resolutions in time, but having the same spatial resolution in each layer. It builds on the block-motion-compensated DCT coding structure of nonscalable video coding in MPEG-2 by adding interlayer prediction between the various layers in scalable coding. Assuming a 2-layer coding scheme consisting of a base-layer and an enhancement-layer, temporal scalability basically involves motion-compensated prediction between the two layers and achieves high coding efficiency. The base-layer signal can be decoded by nonscalable decoders, whereas both layers can be decoded only by temporal scalability decoders.

In Fig. 3 we show a generalized two-layer temporal scalability coding structure (without the preprocessing and postprocessing operations specific to migration to a high-resolution progressive format). Consider, a video signal, in_1, applied to the base-layer encoder and another video signal, in_2, applied to the enhancement-layer encoder. The base-layer encoder is a motion-compensated DCT based nonscalable MPEG-2 encoder, whereas the enhancement-layer encoder is also a motion-compensated DCT encoder with interlayer prediction applied with respect to the base-layer as one of the prediction modes. The encoded bitstreams of base-and enhancement-layers are multiplexed as a single stream in the systems multiplexer. The

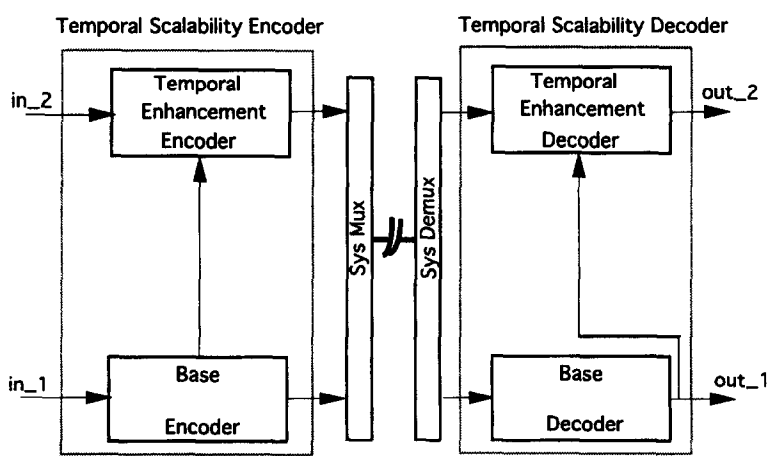

Fig. 3. A simplified codec for temporal scalability.

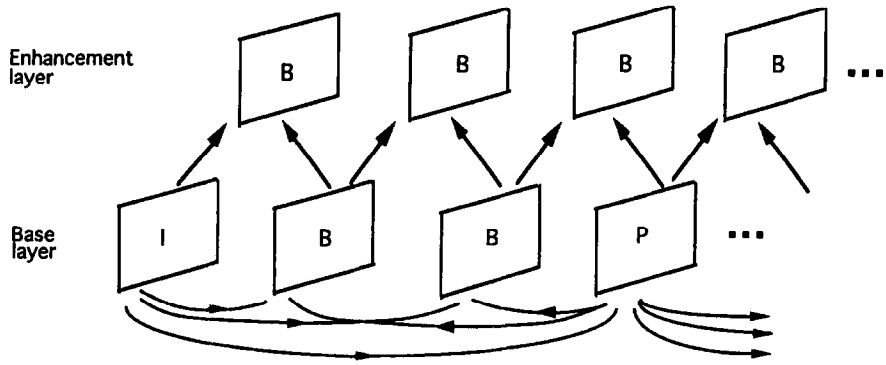

Fig. 4. Bidirectional prediction in enhancement layer of temporal scalability.

systems demultiplexer extracts two bit-streams and inputs corresponding bit-streams to the base and the enhancement layer decoders for decoding, resulting in outputs, out_1 and out_2.

3.2.1. Prediction configuration between layers

Fig. 4 shows an example of temporal scalability configuration with $M = 3$ prediction structure in the base-layer and bidirectional prediction in the enhancement-layer with respect to decoded base-layer frames.

4. Stereoscopic video coding approaches

We are now ready to address the second issue in the realization of practical 3DTV, the issue of efficient representation. We are already aware of the significance of compression in efficient transmission or storage of video and multimedia; the need for video compression becomes even more severe for 3DTV. Stereoscopic 3DTV contains two channels, each carrying a different viewpoint, one intended for left eye, and the other intended for right eye of the viewer. In stereoscopic 3DTV, the raw bandwidth of the video signal is twice that of normal video. The compression of 3DTV video signals is absolutely essential to make 3DTV economically viable. The method of choice is digital compression technology. Digital compression techniques, as in TV and HDTV, provide ease of manipulation, processing flexibility and robustness to errors. Our complete stereoscopic 3DTV system consists of cameras for capturing a scene from two viewpoints, analog-to-digital converters to digitize the video signals, an encoder to compress them, transmission or storage devices for the compressed representation, a decoder and display processor, digital-to-analog converters and finally a stereoscopic display for 3D viewing. In Fig. 5 we show a simplified representation of such a system.

When the system just described is used for broadcast transmission, the compressed digital video signals would need to be modulated prior to actual transmission over-the-air or on cable, and demodulated at the receiver, prior to decompression. However, here our discussion will mainly focus on compression of stereoscopic video signals only, and more particularly on how MPEG-2 video coding techniques can be applied for efficient compression of stereoscopic video.

There are primarily three approaches that can be used for coding of stereoscopic video.

Simulcast stereoscopic coding: This approach involves independent coding of left and right views. Thus, either of the two views may be decoded and displayed as a compatible signal on 2D video receivers.

Compatible stereoscopic coding: In this approach, one view, say the left view, is coded independently; the other view, in this case the right view, is then coded with respect to the independently coded view. Thus, only the independently coded view may be decoded and displayed as the compatible signal.

Joint stereoscopic coding: In this approach, both the left and the right views are coded together.

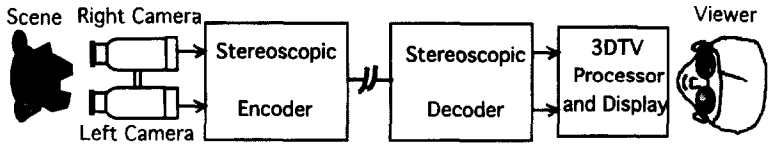

Fig. 5. A stereoscopic TV system.

Since, both views are coded jointly, neither the left view nor the right view may be decoded by itself as a compatible signal. However, there may be exceptions where some restrictions can be imposed allowing a subset signal to be decoded and shown on normal TV as a pseudo-compatible signal.

In the general sense of compression efficiency, the simulcast stereoscopic coding approach is likely to be the least efficient, followed by compatible stereoscopic coding and then the joint stereoscopic coding approaches. This is because simulcast stereoscopic coding independently codes both views, while compatible stereoscopic coding independently encodes only one view and then exploits the correlations between the two views for coding the second dependent view. Likewise, since joint stereoscopic coding does not involve any independent coding at all, in principle, it may allow an even higher compression efficiency, albeit at the cost of lack of compatibility and increased coder complexity.

5. Compatible stereoscopic video coding with MPEG-2

There are several techniques for compatible stereoscopic video coding that are generally an outgrowth of methods that exist for normal video coding. Most of the practical methods are based on motion-compensated DCT coding structure, and although some variations have been reported, the compression performance is believed to be similar. Considering that MPEG-2-based video compression is already either used in or at the verge of being used in a number of applications, it is a reasonable candidate for investigation into coding of stereoscopic video.

5.1. Disparity between views

As discussed earlier, the left and the right views forming stereoscopic video are usually generated under the constraints imposed by the human visual system implying that they are highly correlated. They are generated with camera separations fairly similar to the separation between a pair of human eyes adjusted for the optical magnification of the camera optics. The separation may, at times, be increased to create special effects such as hyperstereo. Thus, the two views of a stereoscopic scene are only slightly different and are easily fused by the human brain to generate the perception of depth. As noted earlier, the difference in perspective between the views is referred to as *disparity*. Stereoscopic video compression attempts to exploit the correlations between the two views by compensating for disparity. Disparity compensation can potentially be more complex and less efficient than motion compensation, although it can be based on similar principles. As an example of disparity values, considering CCIR-601 resolution video and cameras in a convergent configuration, disparity values can be as much as up to 50 pixels or so in the horizontal direction and up to 5 pixels in the vertical direction between left and right frames. However, constraints on the camera set up for stressless viewing of stereoscopic 3DTV and other restrictions tend to reduce the disparity below these extreme levels. For example, with parallel cameras in the same plane, disparity becomes purely horizontal. In this case, epipolar lines become identical to scan lines in the image frames.

5.2. Codec structure based on temporal scalability

The scalability tools of MPEG-2 video are intended for applications where the same video is

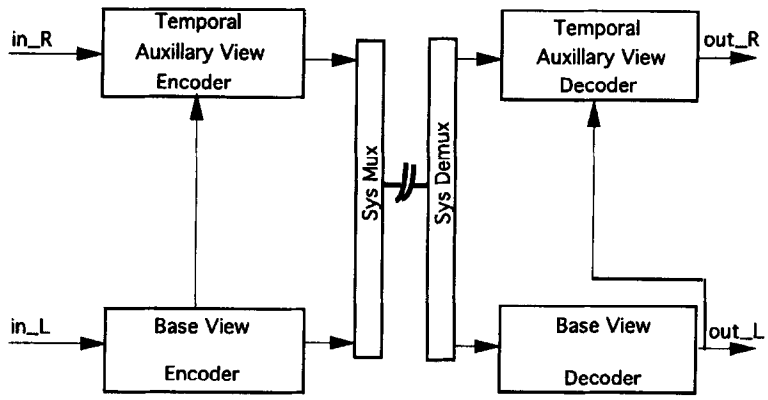

Fig. 6. A high-level structure of a compatible stereoscopic codec.

needed at different resolutions or formats simultaneously. Temporal scalability can be used for coding two video signals of full resolution while exploiting temporal redundancies resulting in very high coding efficiency. In fact, the high-level two-layer stereoscopic codec structure of Fig. 6 is directly derived from the high-level two-layer temporal scalability codec structure of Fig. 3. We are now ready to discuss the operation of various elements of this block diagram.

For stereoscopic video, the left view is input to the base view encoder, a nonscalable MPEG-2 video encoder, and the right view is input to a temporal auxiliary view encoder, an MPEG-2 temporal inter-layer encoder which uses temporal prediction from the decoded lower layer, in this case the decoded left view. Thus, while the left view is coded standalone and provides the compatible signal for normal TV viewing, the right view is encoded with respect to the left view using MPEG-2 temporal scalability. The two resulting bit-streams are packetized by the system multiplexer, Sys Mux, for transmission or storage. The systems demultiplexer, Sys Demux, performs the inverse operation, separating packets representing each layer and feeding unpacketized bit-streams to the base view decoder and the temporal auxiliary view decoder, which are a nonscalable MPEG-2 video decoder and MPEG-2 temporal inter-layer decoder, respectively. Although we have chosen to code the left view independently and the right view dependently, if needed the structure could easily be reversed such that the right view would be the independently coded signal and the left view the dependently coded signal. For the purposes of illustration, we will continue with the use of the left view as the independently coded signal and the right view as dependently coded signal.

5.3. Disparity-compensated prediction

A little earlier, in our discussion about disparity we noted the similarities, at least in principle, between disparity compensation and motion-compensation. In stereoscopic video, the left view pictures and the right view pictures are assumed to be independently generated; this has the implication that for each right view picture there is a corresponding left view picture, coincident in time. Of course, each of the corresponding left view and right view pictures, although they are taken at the same instance in time, are different depending on the amount of disparity in the scene.

Since nonscalable MPEG-2 coding is quite efficient due to bidirectional motion-compensated prediction in B-pictures, and also, since temporal scalability is highly efficient mainly due to the use of bidirectional prediction in B-pictures, it is conceivable that high coding efficiency may also be obtained by coding the right view pictures as B-pictures using bidirectional prediction. The bidirectional prediction employed here uses two

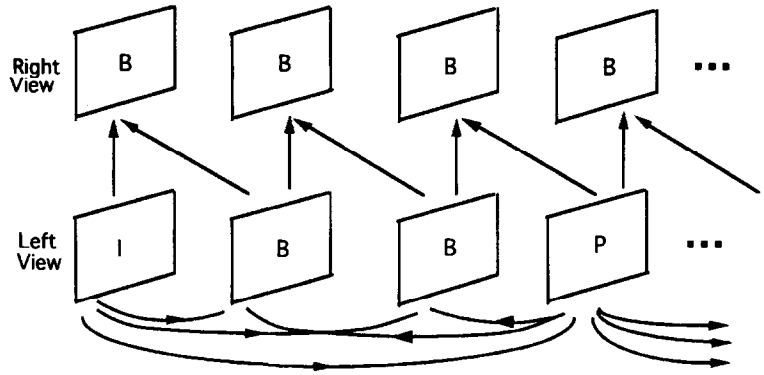

Fig. 7. Disparity-compensated prediction configuration.

disparity-based predictions; the resulting prediction configuration is shown in Fig. 7.

The left view may be coded with nonscalable MPEG-2 coding. As an example, one can employ I-pictures, P-pictures and B-pictures, arranged in an $M=3$ structure of Fig. 2 quite popular in MPEG-1 and MPEG-2 video encoding. As a reminder, I-pictures do not use any prediction, P-pictures use forward prediction, and B-pictures use bidirectional prediction employing two different kinds of prediction, forward and backward prediction, which result in three different modes, forward, backward and averaged prediction. Furthermore, in using frame-picture coding, $M=3$ structure means that there are two B-frame pictures between a pair of reference frame pictures, each of the reference frames can be an I-frame picture or a B-frame picture.

As mentioned earlier, the right view would be coded with disparity-compensated coding (derived as an adaptation of MPEG-2 temporal inter-layer coding) with predictions with respect to two decoded left view frames. More specifically, it employs B-pictures, as they use bidirectional prediction, consisting of two different kinds of prediction, both referencing decoded left view frames. The two reference frames used for prediction are the left view frame coincident in time with the right view frame to be predicted, and the other is the left view frame temporally next in display order to the right view frame being predicted. As before, the two predictions result in three different modes called forward, backward and averaged prediction. In this context, due to interlayer predictions, the terminology of forward versus backward gets confusing. Here, we use forward prediction to signify prediction from the coincident left view frame and backward prediction to mean prediction from the left view frame next in display order to the coincident frame. Since prediction of a right view frame is done by two left view frames, this method attempts to predict disparity between the views and is called disparity compensated prediction. It requires the calculation of disparity vectors at the encoder that are included in the right view coded bit-stream. The right view decoder uses the disparity vectors to build disparity-compensated bidirectional predictions. It should be expected that the backward prediction mode will tend to produce larger prediction errors than the forward or averaged modes when there is motion in the scene due to the lack of temporal coincidence.

5.4. Disparity- and motion-compensated prediction

This prediction configuration also uses bidirectional prediction for the same reasons of high prediction efficiency as nonscalable MPEG-2 video coding with B-pictures and coding using temporal scalability with B-pictures. However, the bi-directional prediction here involves a disparity-based prediction and a motion-based prediction within the right view sequence. The resulting prediction configuration is shown in Fig. 8.

The coding of the left view uses the same nonscalable MPEG-2 coding as described for the

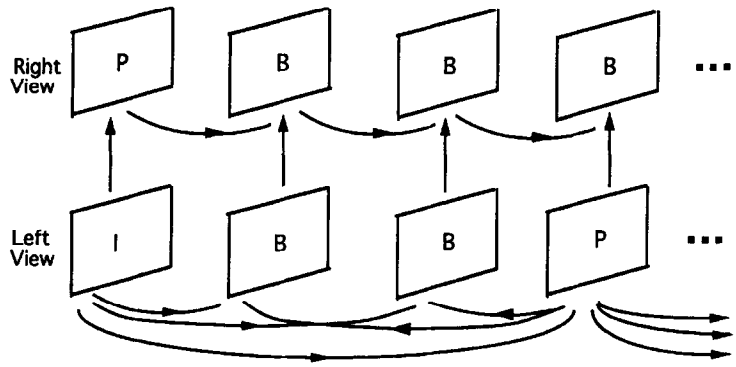

Fig. 8. Disparity- and motion-compensated prediction configuration.

disparity-compensated prediction configuration and will not be discussed again here.

The right view coding uses disparity- and motion-compensated coding (derived as an adaptation of MPEG-2 temporal inter-layer coding) with predictions with respect to a decoded left view frame and a decoded right view frame. The two reference frames used for prediction are the left view frame coincident with the right view frame to be predicted and the previously decoded right view frame in display order. As before, the two predictions result in three different modes called forward, backward and averaged prediction. Again, due to inter-layer prediction, the terminology of forward versus backward gets confusing. Here, we use forward prediction to signify prediction from a previous decoded right view frame in display order and backward prediction to mean prediction from the left view frame coincident with the right view frame being predicted. Since one prediction is done with respect to the left view, this is a disparity-compensated prediction, where as in the second prediction, a right view frame is predicted from a previously decoded right view frame, this prediction is simply a motion-compensated prediction. Both disparity vectors and motion vectors needed for bi-directional prediction of the right view are calculated at the encoder and included in the right view coded bit-stream. The right view decoder computes disparity and motion-compensated bi-directional prediction from these vectors.

Perhaps a comment about the first frame of the right view is in order. It is coded as a P-frame picture and uses only a coincident left view frame for prediction. In all generality, P-frames can occur whenever desired in the coding of the right view. Prediction of the first right frame can also use the left view frame next to the temporally coincidental frame in display order. Again, it is worth noting that the specific prediction configuration just described can have other variations also supported by the MPEG-2 temporal scalability syntax.

5.5. Stereoscopic codec structure

Now that we have a somewhat clear understanding of the prediction configurations to use, it is time to revisit the high-level codec structure of Fig. 6 and derive a more precise structure of a compatible stereoscopic codec. We have already indicated that the base-view encoder and the base-view decoder are simply a nonscalable MPEG-2 encoder and decoder, respectively. For clarity, we rename them the motion-compensated DCT encoder and the motion-compensated DCT decoder, to make explicit the coding method used. Also, we mentioned that the temporal auxiliary view encoder performs the function of temporal inter-layer encoder, using prediction with respect to decoded base view frames. Based on our discussion of prediction configurations, in the generalized case, both disparity and motion-compensated prediction are employed. Considering this and the renaming convention used for the base-view, the temporal auxiliary view encoder and decoder are renamed the disparity and

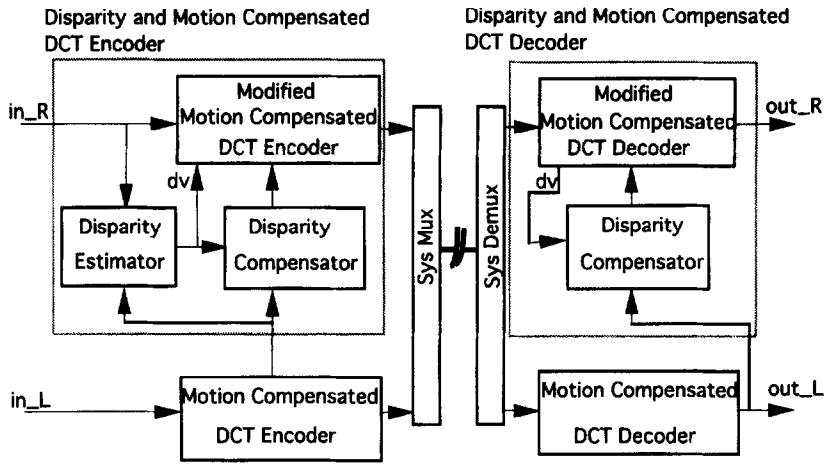

Fig. 9. A high-level stereoscopic codec with a few details.

motion-compensated DCT encoder and decoder, respectively. Furthermore, disparity compensated encoding requires a disparity estimator and a disparity compensator, pretty much the same way that motion-compensated encoding requires a motion estimator and a motion compensator. On the other hand, disparity compensated decoding only requires a disparity compensator. The resulting codec structure with renamed encoders and decoders as well as more details is shown in Fig. 9. Also, in this figure, disparity vectors are illustrated by dv.

The motion-compensated encoder included in the disparity- and motion-compensated DCT encoder is so named because it employs modified B-pictures which can use motion- and disparity-compensated predictions. Disparity vectors are generated by the encoder and are used for disparity compensation both at the encoder and decoder. These vectors along with motion vectors, coded DCT coefficients and overhead indicating specific motion and coding modes form the bit-stream that is sent to the corresponding decoder. Each of the aforementioned encoders and decoders produces disparity and motion-compensated predictions using decoded frames and thus remain synchronized in their feedback loops, in the absence of channel errors. The operation of SysMux and SysDemux is exactly the same as described earlier.

With frame-pictures, MPEG-2 motion estimation uses 16×16 frame-blocks or 16×8 field-blocks. The coding first involves taking the DCT of either 8×8 frame or field-blocks of a picture obtained by motion-compensated prediction. Macroblock based selection is then used to determine the best DCT coding mode. It is worthwhile pointing out that disparity estimation and compensation can be performed in exactly the same way as motion estimation and compensation, i.e., with half-pixel accuracy on 16×16 frame blocks or 16×8 field-blocks which ever gives better results on a macroblock basis. The primary difference is that disparity estimation and compensation is performed between coincident frames of the left view and right view, and that for block matching estimation, it uses a much larger range for the horizontal search as compared to vertical search due to the geometry of stereoscopic imaging and display.

6. Improvements by gain and offset correction

The acquisition of stereoscopic video in most cases requires that separate cameras be used to acquire the left and right views. This introduces the possibility of mismatches between the camera outputs. These mismatches may make it difficult for viewers to merge the images or cause visual fatigue. They can also reduce the efficiency of the coding. In an ideal case, the complete electro-optical transfer function of the two cameras should be matched

before acquiring stereoscopic video. However, in the real world, a codec must be able to deal with video from mismatched cameras in real time without the benefit of a known reference. The best that can be done in this case is to attempt to adjust the transfer function of one view to give a good visual match to the other view.

Adequate compensation can usually be obtained with simple gain and offset adjustments applied to the luminance and chrominance components of one of the views. Since we have assumed that the left view is coded independently, it can be used as the reference for computation of luminance and chrominance mismatch parameters and the right view can be compensated for this mismatch. This type of compensation can be applied globally, on an entire image basis, or locally, for each block of pixels. Both local and global compensation have some advantages and disadvantages. For example, global methods may use matching of histograms, whereas local methods may use minimization of mean square error between blocks of the right image and decoded left image. Global compensation has the advantage that at most 6 parameters must be transmitted per frame. In block compensation, 6 parameters must be transmitted per block and the coding overhead could be significantly increased. Block compensation also introduces the possibility of increasing the visibility of block boundaries. For this reason, block compensation would only be used to increase coding efficiency and it would have to be removed at the decoder. Thus, block compensation would do nothing to improve the visual match between the left and right images.

Since our main interest is in increasing compression efficiency when using disparity-compensated coding, mismatch correction during pre-processing or during coding can be used. However, only the preprocessing approach is entirely compatible with the disparity- and motion-compensated coding approach that we have discussed based on MPEG-2.

The technique of luminance and chrominance gain-mismatch compensation can be briefly outlined as follows. Let $S_L(x, y)$ represent the luminance or chrominance value of a pixel located at (x, y) in the left view and $S_R(x, y)$ represent the corresponding luminance or chrominance intensity value of a pixel located at (x, y) in right view. Then, assuming a linear model using a gain a and offset of b, $S'_R(x, y)$, a corrected value can be written as follows:

$$S'_R(x, y) = aS_R(x, y) + b.$$

Now, using real image intensities in a given stereoscopic scene, it becomes possible to mathematically compute the values of a and b for that scene by minimizing the difference between $S'_R(x, y)$ and $S_L(x, y)$, either globally on a picture basis or locally on a block basis. Once the values of a and b are known, they are substituted into the same equation to compute the mismatch corrected values of the right view of that scene. Of course, three such parameter computations are necessary, one for luminance and one each for the two chrominance signals.

6.1. Gain and offset estimation and correction

Histograms of the image data provide a convenient representation for the computation of the gain and offset parameters. Any two features present in both the right and left histograms can be used. These features could be two prominent peaks or a peak and an end point. The problem then is one of automatically identifying suitable features in the preprocessor. This in itself is a difficult problem which may not be solvable for a significant range of video input.

A more suitable method for gain and offset estimation between the left and the right views involves matching global geometric properties of the left and right view histograms. This implicitly minimizes mismatch error. To determine the gain, a, and the offset, b, by this method we require that the first and second moments of the histograms of the left view and gain and offset corrected right view be equal.

Equating the first moments we have

$$E(x_L) = E(ax_R + b), \tag{1}$$

where the expectation is $E(x) = \sum_{i=1}^{N-1} n_i x_i$, with n_i being the fraction of pixels with value x_i and N the number of levels.

From Eq. (1), the offset, b, can be rewritten as

$$b = E(x_L) - aE(x_R). \tag{2}$$

The E terms in Eq. (2) can be replaced by appropriate statistical first moments, I_{1L} or I_{1R}; with this substitution, the following equations result:

$$b = I_{1L} - aI_{1R}. \tag{3}$$

Equating the second moments we have

$$E(x_L^2) = E([ax_R + b]^2). \tag{4}$$

Expanding Eq. (4) and substituting b from Eq. (2) we have

$$E(x_L^2) = a^2E(x_R^2) + 2aE(x_R)[E(x_L) - aE(x_R)] + [E(x_L) - aE(x_R)]^2. \tag{5}$$

Solving Eq. (5) results in the following:

$$E(x_L^2) = a^2E(x_R^2) + (E(x_L))^2 - a^2(E(x_R))^2. \tag{6}$$

Now rewriting Eq. (6) we derive, gain, a:

$$a = \sqrt{\frac{E(x_L^2) - (E(x_L))^2}{E(x_R^2) - (E(x_R))^2}}. \tag{7}$$

Again, replacing the expectation, E, with the appropriate statistical moments in Eq. (7) we get

$$a = \sqrt{\frac{I_{2L} - (I_{1L})^2}{I_{2R} - (I_{1R})^2}}, \tag{8}$$

where I_{2L} is the second moment of the left view and I_{2R} is the second moment of the right view.

Finally, the gain and offset corrected right view can be calculated for each pixel as follows:

$$x_{RC} = ax_R + b. \tag{9}$$

The gain and offset parameters obtained by this procedure are only accurate if the content of the left and right views are essentially identical. Disparity between the views may make an object visible in one view and not in the other. This could distort one of the histograms in such a manner that the method described above does not yield a good overall correction. One possible way around this problem is to use the prediction error to determine if the calculated gain and offset are valid and only apply the correction if it does not adversely affect the prediction error. Also, to avoid strobing effects as the gain and offset attempt to follow the scene content, the gain and offset parameters should probably only be changed at scene cuts.

6.2. Stereoscopic video codec with gain and offset correction

Fig. 10 shows the high-level structure of a compatible codec for stereoscopic video discussed earlier along with the gain and offset mismatch correction just described. The left view is coded

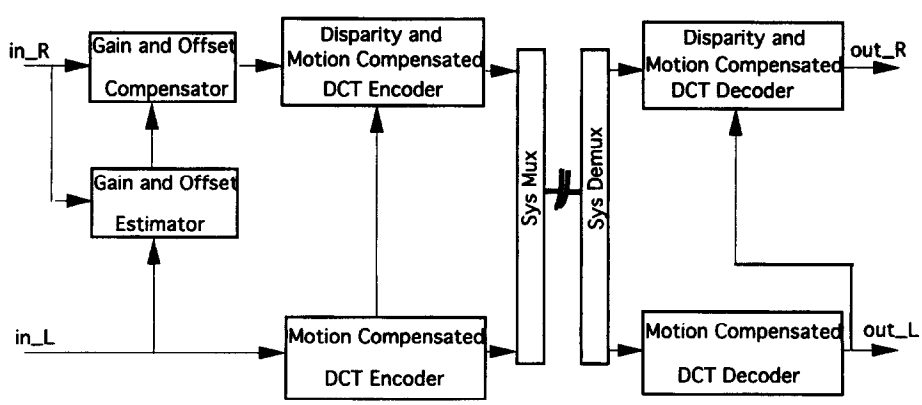

Fig. 10. A stereoscopic codec with global gain and offset correction preprocessing.

independently by a motion-compensated DCT encoder as before. However, the right view is compared in the gain estimator and compensator preprocessor with the left view to estimate its luminance and chrominance gains and offsets to produce a modified right view signal. This modified right view signal is then fed to the disparity- and motion-compensated encoder, which also uses decoded left view signals for prediction. It may also be necessary to provide a feedback path (not shown) from the disparity- and motion-compensated DCT encoder to the gain compensator to turn off the compensation if the coding efficiency decreases. The details of the operation of the two encoders, SysMux, SysDemux and the corresponding decoders is exactly the same as that described for Figs. 3 and 6.

7. Stereoscopic video coding for MPEG-4

Besides, MPEG-2 which now already includes coding of stereoscopic video (for which the subjective quality evaluation tests are in progress), the ongoing MPEG-4 standard is also expected to address coding of stereoscopic video. In general, the MPEG-4 standard is addressing a new generation of functionalities [1, 8] such as content-based interactivity, high compression and universal access. Content-based interactivity includes functionalities such as content-based multimedia data access, content-based manipulation, hybrid natural and synthetic data and improved temporal random access. High compression includes functionalities of improved coding efficiency for normal video as well as efficient coding of multiple concurrent data streams; stereoscopic video is one example of multiple concurrent data streams. Universal accessibility includes robustness in error-prone environments and content-based scalability. The MPEG-4 standard, while it is expected to operate over a range of bit-rates is being optimized at low bit-rates. The MPEG-4 standard is also different from MPEG-1 and MPEG-2 in the sense that it is expected to offer a higher degree of freedom in the use of coding tools by using a flexible syntax. It is anticipated that a range of coding tools from traditional motion-compensated block-based coding to region-based, object-based and model-based coding may be included in MPEG-4 for specialized functionalities.

As mentioned earlier, multiple concurrent data-streams functionality in MPEG-4 is likely to support coding of 2 or more simultaneous views; when we deal with more than 2 views, we will call it multi-viewpoint coding (more on this later). Even before MPEG-4 decided to include functionality of high-efficiency coding of stereoscopic and multi-viewpoint video, various researchers [23, 25] have already been investigating these issues.

The schedule of MPEG-4 development called for submission of complete algorithms and tools for first evaluation which was conducted in November 1995. Stereoscopic video coding was not formally tested in November 1995 tests, although submitted proposals were informally evaluated based on coded results and documentation. The second evaluation which is expected to be conducted in July 1997 may involve formal testing of stereoscopic video coding algorithm developed. The MPEG-4 standard is expected to achieve a stable stage by November 1997 and be finalized by November 1998.

7.1. MPEG-4 bit-rates and sequences

MPEG-4 finalized two stereoscopic scenes of 10 s each called 'Train and Tunnel' and 'Funfair' for its evaluations. It also finalized two bit-rates, 512 kbit/s and 1024 kbit/s, both for coding of left view as base-layer. The bit-rates for coding of right view in each case were left unspecified. The source sequences were provided at CCIR-601 4:2:2 resolution and format. No constraints were imposed on the spatial and temporal resolutions and formats to be employed in coding.

7.2. Proposed generalized codec

Our proposal for coding of stereoscopic video in MPEG-4 was designed based on the proven framework of stereoscopic coding for MPEG-2, to which we contributed temporal scalability based solution described earlier in the paper. However,

for MPEG-4 stereoscopic applications we were faced with significantly lower bit-rates, choice of a variety of picture formats and frame rates, and maintaining meaningful bit-rate ratios between coded left view and right view. In addition, for reasonable quality stereoscopic viewing, very low frame rates are not considered acceptable so higher frame rates have to be coded. Furthermore, since in MPEG-4 only two sequences were chosen for evaluation and the peculiarities of stereoscopic imaging can result in unanticipated difficulties in compression, it was decided that the codec structure allow ease of incorporation of preprocessing, postprocessing and in loop processing that would be absolutely necessary in some cases.

The proposed codec performs compatible stereoscopic coding and is an extension of the SBASIC codec proposal [17] submitted to MPEG-4 (for November 1995 tests) for coding of generic video scenes at low bit-rates of 320, 512 and 1024 kbit/s. SBASIC stands for Scene and Bandwidth Adaptive Scalable Interframe Coding. We now show details of the proposed stereoscopic codec in Fig. 11 and describe it next.

The main components of the codec of Fig. 11 are scene and bandwidth adaptive preprocessor, temporal rate adaptive postprocessor, spatial upsampler, global mismatch estimator and global mismatch corrector, MPEG-2++ encoder/decoder and modified temporal auxillary view encoder/decoder.

The left and right views are input to scene and bandwidth adaptive preprocessor which has the primary responsibility of generating the video resolution and formats commensurate with the total bandwidth available and a variety of other conditions. This preprocessor accepts several control signals (*gen par0*) and outputs other control signals (*cod par0*), some of which are utilized by the MPEG-2++ Encoder/Decoder used for coding of left view and modified temporal auxillary encoder/decoder used for coding of the right view. The control signals at the input and output of MPEG-2++ encoder are *cod par1* and *gen par1*. Similarly, the control signals at the input and output of modified temporal auxillary view encoder are *cod par2* and *gen par2*. The corresponding decoders employ control signals *cod par1* and *cod par2* and are derived from coded bit-stream. The control signals *cod par1* and *cod par2* are also intended for temporal rate adaptive postprocessor. The upsampler currently employed simply allows decoded frames to pass through without any spatial conversions, although, for example, we expect it to provide 1:2 upsampling so that the right view may be coded at one-half the resolution of left view, in future.

The left-view coder is syntactically the same as that of MPEG-2 coder albeit with modified semantics.

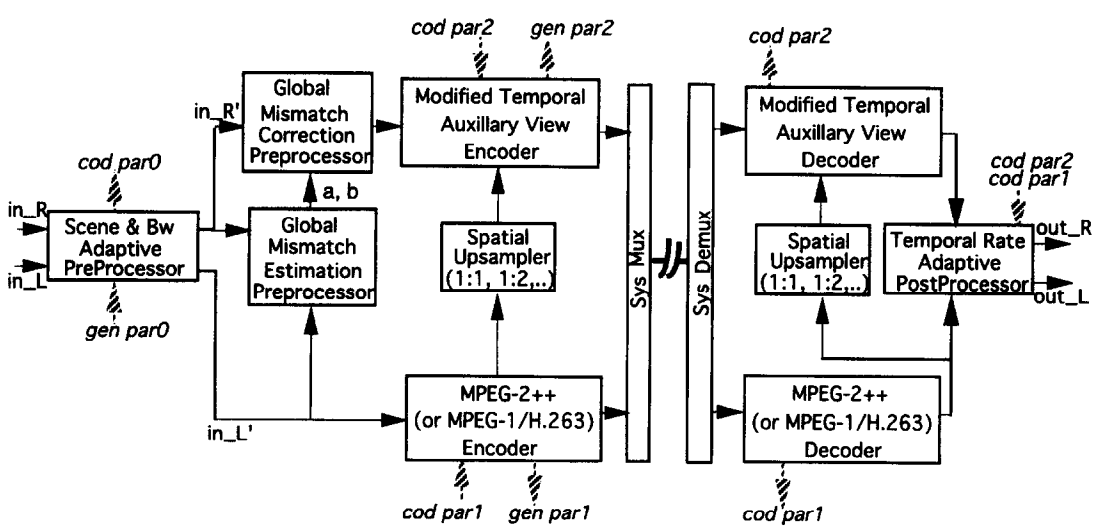

Fig. 11. Proposed generalized codec structure for MPEG-4 stereoscopic video coding.

The left-view coder is itself assumed to be temporally scalable to allow better control of the limited allocated bit-rate. Likewise, the right-view coder is also syntactically the same as that of MPEG-2 temporal enhancement coder but the semantics are modified. The right-view coder mainly codes B-pictures; however, I- and P-pictures can also be coded if necessary. The three main prediction modes used with right-view B-pictures are motion-compensated prediction, disparity compensated prediction and an average of disparity and motion-compensated prediction. The disparity estimates are computed in the same manner as motion estimates with the difference that they are computed between the simultaneously occurring frames (on a block basis) of the decoded left view and blocks of right view. Further, if necessary conditions for stereoscopic imaging are satisfied, disparity is expected to mainly horizontal only. Thus, large disparity vector range is employed horizontally, while only a small range is employed vertically.

The two outputs of scene and bandwidth adaptive preprocessor, in_L′ and in_R′ are intended for the left-view encoder and the right-view encoder, respectively. We use left view as the reference and present both the left view and the right view to global mismatch estimator which determines 2 mismatch parameters per component (Y, Cr, Cb); these values are referred to as gain, a, and offset, b. The gain and offset values just computed are then sent to global mismatch corrector which also takes as its input the right-view signal output by scene and bandwidth adaptive preprocessor. After correction, the mismatch corrected right view appears at the output of global gain and offset correction preprocessor and is fed to modified temporal auxillary encoder used for coding the right view and the left view is input to MPEG-2++ encoder used for coding the left view. In the proposal submitted to MPEG-4 for November 1995 evaluation, mismatch correction was the same as the gain and offset preprocessing described in Section 6.

7.3. Mismatch correction

Other than mismatches due to differences due to disparity (and in some cases due to motion) between left and right views of a stereoscopic scene, a number of other mismatches can occur during imaging. In the two imaged views of a scene, these mismatches may appear due to dissimilar transfer function of cameras, dissimilar light intensities, dissimilar imaged area size, differences in perspective and others. Among the types of corrections that can be applied to correct for these mismatches are luminance and chrominance gain and offset correction, and correction of perspective.

In Section 6, we have already discussed gain and offset correction; other researchers [2] have partially tried correcting for perspective differences. As mentioned earlier, in our proposal [11] to MPEG-4 we have only employed gain and offset correction; however, we consider other types of mismatch correction to be important and is currently the subject of ongoing study [7, 12].

7.4. Coding details

In our proposal for stereoscopic coding for MPEG-4 we have employed bit-rates of 1024 kbit/s for left view and targeted coding of right view in 512 kbit/s. Since stereoscopic coding for display with good quality requires full field rate of 60 fields/s (50 fields/s in European format), we retain full resolution for luminance signal vertically as is available in CCIR-601 4:2:0 signal. For chrominance signals we also retain the vertical resolution of CCIR-601 4:2:0 which is usually derived by decimation of CCIR-601 4:2:2 chroma signal by a factor of 2 vertically. However, since the bit-rates for MPEG-4 stereoscopic video coding are very challenging we further decimate luminance and chrominance components of CCIR-601 4:2:0 signal horizontally by a factor of 2 to derive the half-horizontal resolution (HHR) format. Thus, our left view and right view are coded at HHR resolution. For display purposes, the decoded HHR luminance and chrominance components are upsampled to produce CCIR-601 4:2:2 resolution. The filters used in downsampling and upsampling to derive the HHR format are MPEG-2 standard filters [21].

The left view of HHR resolution is coded as frame pictures using the motion compensated

prediction (with 16×16 frame, 16×8 field and dual-prime modes) and DCT coding (with 8×8 frame and field modes) as described in Section 3.1. For coding of test sequences chosen by MPEG-4, in coding of left view, no B-pictures were employed ($M = 1$) to allow better rate control at these low bit-rates; under aforementioned conditions dual-prime mode was found to perform comparably to B-pictures. For other experiments (not reported here) we have not only used B-pictures, but in fact temporal scalability for coding of left-view itself as two layers. To meet the target bit-rate a number of other actions were necessary such as, fairly steep inter- and intra- quantization matrices, reduced frequency of I-pictures, etc.

The right view of HHR resolution after gain and offset correction is coded using temporal scalability B-pictures with disparity- and motion-compensated configuration. Restrictions are placed on the range used in disparity estimation, this in turn also has the benefit of reducing number of bits used for coding of disparity vectors. Motion estimation and compensation of temporal scalability B-pictures uses two types of predictions as explained for the case of MPEG-2 based stereoscopic coding. The prediction modes employed for forming disparity- and motion-based predictions are 16×16 frame and 16×16 field modes. DCT coding employs 8×8 frame or field blocks as for left view. Due to severe bit-rate limitations in coding of right view, extremely steep quantization matrices are employed and often result in retaining mainly the low-frequency coefficients.

8. Coding results

We illustrate the performance of MPEG-2 video coding on four stereoscopic video scenes available in European CCIR-601 4:2:2 format of 720 pixels/line and 576 lines/frames at rate of 25 interlaced frames/s. These test sequences are 'Train', 'Manege', 'Tunnel' and 'Aqua' and each sequence is available as two views, a left view and a right view. For coding, sequences are first reduced to 4:2:0 format by vertical subsampling the chrominance signals by a factor of 2. Each sequence is then coded using the two prediction configurations discussed earlier. One uses disparity compensated prediction only and the other uses both motion and disparity compensated predictions. The left view is coded independently by using nonscalable MPEG-2 coding using frame-pictures with a prediction distance of $M = 3$ (two B-pictures) between pairs of reference pictures, and an Intra picture distance of $N = 12$. The right view is coded dependently with respect to the decoded left view frames using modified B-pictures with a coding structure dependent on the specific prediction configuration employed. We then illustrate the performance of our proposed coding technique for MPEG-4 using two long sequences of 10 s each; these sequences are extended versions of the two sequences used for MPEG-2 coding. For coding at MPEG-4 bit-rates HHR resolution is employed which is essentially half-horizontal resolution of CCIR-601 4:2:0, both for luminance and chrominance signals.

In stereoscopic video coding, it is not necessary to code both views with the same quality; in most cases, the quality of one of the views can be significantly lower compared to the other view without incurring significant visual distortions. This fact is supported by several practical experiments that claim that the quality of one view can be lower by 3 dB or more as compared to the other view. Further support for unequal distribution of quality between the two views is provided by the leading-eye theory, according to which the overall subjective impression in stereoscopic 3D viewing depends mainly on the quality perceived by the leading eye of viewers. For a given limited bit-rate, significant benefits can accrue from unequal distribution of bits between the two views, and thus this approach is adopted frequently. In our discussion, the left view is coded independently and is intended to be the compatible signal for normal TV. The left view is coded with high quality, say at 4 or 6 Mbit/s. The right view is coded dependently, is intended to have lower quality, and thus requires a lower bit-rate. It can be coded at significantly lower bit-rate of around 2 Mbit/s.

In Table 1 we show SNR results of independent coding of the left view using MPEG-2 nonscalable coding with frame-pictures, $M = 3$ and $N = 12$.

In Table 2, SNR results are compared for coding of the right view using disparity compensation

Table 1
Luminance SNR (dB) for left view coded at 4 Mbit/s using nonscalable MPEG-2 with $N = 12$ and $M = 3$

Sequence	Left view
Train	35.81
Manege	29.85
Tunnel	35.91
Aqua	35.38

Table 2
Luminance SNR (dB) for right view coded at 2 Mbit/s with MPEG-2 (left view coded at 4 Mbit/s with nonscalable MPEG-2)

	Right view at 2 Mbit/s (Left view at 4 Mbit/s)		
Sequence	Simulcast coded	Disparity-comp. coded	Disparity- and motion-comp. coded
Train	31.75	33.13 (+ 1.38)	34.08 (+ 2.33)
Manege	26.64	26.19 (− 0.45)	28.67 (+ 2.03)
Tunnel	33.12	29.84 (− 3.28)	34.15 (+ 1.03)
Aqua	30.59	27.54 (− 3.05)	30.84 (+ 0.25)

(disparity comp.) and disparity and motion compensation (disparity and motion comp.) with Simulcast coding. All results are at 2 Mbit/s.

The results in Table 2 indicate that with the chosen bit partitioning between left and right views, disparity- and motion-compensated coding works significantly better by anywhere from 0.25 to about 2.25 dB when compared to simulcast. This table also shows that simulcast, in some cases significantly outperforms disparity compensated coding by as much as about 3 dB. In other cases, disparity only compensated coding either performs the same as simulcast or outperforms it. Thus, it can be concluded that disparity- and motion-compensated coding provides the best overall performance, whereas disparity compensated coding falls short of expectations. As a reminder, both of these techniques use B-pictures which use bi-directional prediction resulting in three basic prediction methods.

The results discussed here are based on stereoscopic video coding using MPEG-2 temporal scalability [18]. They are among the more promising results using any block based compatible stereoscopic video coding scheme. Experiments using various MPEG-2 scalabilities for coding of stereoscopic video have also been reported in [22] and further experiments using temporal scalability have also been reported in [20]. Other compatible approaches in the past have either used disparity-compensated coding or disparity- and motion-compensated coding [4]. One possible reason for the good performance of the MPEG-2 temporal scalability approach is that it uses an interpolated prediction mode which uses averaged prediction. When either disparity or motion compensation by themselves do not perform adequately, the interpolation mode still performs well. The frequency of selection of various prediction modes is compared in Table 3.

As Table 3 shows, in disparity-compensated coding the percentage of times the interpolation mode

Table 3
Percentage of right view coded blocks in different prediction modes for disparity-compensated and motion and disparity-compensated configurations in Table 2

	Prediction mode with right view disparity-comp. coded			Prediction mode with right view disparity and motion comp. coded		
Sequence	% Forward	% Backward	% Averaged	% Motion	% Disparity	% Averaged
Train	18.2	9.7	70.5	25.5	14.4	58.6
Manege	20.8	14.7	55.0	60.1	2.1	32.9
Tunnel	19.8	11.3	63.5	50.7	1.6	30.9
Aqua	12.1	15.0	68.2	47.7	5.0	42.1

is used is about 65% on the average. This indicates that, quite often, each of the disparity estimates cannot provide good prediction and thus the interpolation mode is needed. Also, each of the two disparity estimates are selected nearly as frequently. The only sequence in which backward prediction using disparity compensation is comparable to forward prediction is Aqua. This sequence has very little motion and backward prediction is not greatly affected by the lack of temporal coincidence. This table also shows that in disparity- and motion-compensated coding, the interpolation mode is selected about 40% on average. Motion-compensated prediction is selected at least as often as interpolation and more than disparity compensated prediction. Overall, when compared to simulcast coding, which uses only motion-compensated coding, the higher performance of disparity- and motion-compensated coding can be attributed to use of the interpolated mode which uses both motion- and disparity-compensated prediction.

Next, in Tables 4 and 5 we confirm the performance of MPEG-2 temporal scalability for stereoscopic video coding using a bit assignment of 6 Mbit/s for left view and 2 Mbit/s for right view, suitable for higher-quality applications.

Table 5 shows that the simulcast coding outperforms disparity compensated coding in most cases. Furthermore, as for the case of coding the left view at 4 Mbit/s and the right view at 2 Mbit/s, in all cases, the best performance for right-view coding is obtained by combined motion and disparity based coding.

Table 4
Luminance SNR (dB) for left-view coded at 6 Mbit/s using nonscalable MPEG-2 with $N = 12$ and $M = 3$

Sequence	Left view
Train	37.70
Manege	31.91
Tunnel	37.55
Aqua	37.45

Table 5
Luminance SNR (dB) for right-view coded at 2 Mbit/s with MPEG-2 (left view coded at 6 Mbit/s with nonscalable MPEG-2)

	Right view at 2 Mbit/s (Left view at 4 Mbit/s)		
Sequence	Simulcast coded	Disparity-comp. coded	Disparity- and motion-comp. coded
Train	31.75	33.58 (+ 1.83)	34.56 (+ 2.81)
Manege	26.64	26.36 (− 0.28)	28.84 (+ 2.20)
Tunnel	33.12	29.95 (− 3.17)	34.36 (+ 1.24)
Aqua	30.59	27.78 (− 2.81)	31.19 (+ 0.50)

The frequency of selection of the various prediction modes for the case of 6 Mbit/s for the left view and 2 Mbit/s for the right view are shown in Table 6.

Overall, the results at 6 Mbits/s are very similar to those at 4 Mbits/s and the conclusions from Table 6 are the same as those from Table 3.

Table 6
Percentage of right-view coded blocks in different prediction modes for disparity compensated and disparity and motion-compensated configurations in Table 5

	Prediction mode with right view disparity-comp. coded			Prediction mode with right view disparity- and motion-comp. coded		
Sequence	% Forward	% Backward	% Averaged	% Motion	% Disparity	% Averaged
Train	18.5	8.8	71.6	24.1	15.2	59.0
Manege	20.6	14.9	55.5	58.6	2.4	34.1
Tunnel	16.4	9.8	68.1	50.4	1.7	30.9
Aqua	11.9	13.5	69.4	46.6	5.8	42.5

Table 7
Luminance and chrominance SNR (dB) of the right view coded at 2 Mbit/s with and without gain and offset corrected preprocessing; left view is coded at 6 Mbit/s with nonscalable MPEG-2

	Right view disparity- and motion-comp. coded			Gain corrected preprocessed right view disparity- and motion-comp. coded		
Sequence	Y	Cb	Cr	Y	Cb	Cr
Train	34.56	36.99	37.36	34.97 (+ 0.41)	38.12 (+ 1.13)	39.24 (+ 1.88)
Manege	28.84	35.37	35.47	29.70 (+ 0.86)	35.30 (− 0.07)	35.35 (− 0.12)
Tunnel	34.36	37.98	38.96	34.37 (+ 0.01)	38.05 (+ 0.07)	39.16 (+ 0.20)
Aqua	31.19	36.65	37.85	31.40 (+ 0.21)	37.00 (+ 0.35)	38.35 (+ 0.50)

We now compare in Table 7 the coding performance for the right view with and without gain and offset-compensated preprocessing. Since gain correction is applied not only to luminance but also to chrominance signals, the SNR statistics for luminance as well as chrominance signals is compared. The global gain and offset parameters were determined by the method described in Section 6.1.

The improvement due to gain and offset correction obviously varies depending on the amount of mismatch between the left and right views of each stereoscopic scene. In Table 8 we observe that the improvement in luminance SNR is most significant for the 'Manege' sequence, whereas the improvement in chrominance SNR is most significant for the 'Train' sequence. In SNR sense, the improvements in luminance appear least significant for 'Tunnel' followed by 'Aqua' sequence; the improvements in chrominance appear least significant for 'Manege' followed by the 'Tunnel' sequence. In all cases, the improvement in quality due to gain and offset correction was noticeable, although it was most striking for the 'Train' and the 'Manege' sequence. For 'Manege' sequence at CCIR-601 4:2:0 resolution, Fig. 12 shows original frames as well as coded frames without and with gain and offset corrected preprocessing. Fig. 13 shows similar results for the 'Tunnel' sequence.

Table 8
SNR for left view of HHR resolution coded at 1 Mbit/s with $M = 1$ by our MPEG-4 proposal

Sequence	Left view
Train&Tunnel	31.39
Funfair	24.70

We now present results of our stereoscopic video coding proposal for MPEG-4; the main goal of the proposal is to allow coding with reasonable quality at significantly lower bit-rates. The bit-rates targeted are 1 Mbit/s for the left view and about 0.5 Mbit/s for the right view. For these experiments, due to challenging bit-rates, lower resolution of HHR 4:2:0 (rather than CCIR-601 4:2:0) is used for coding. However, for display, HHR 4:2:0 is upsampled to full CCIRR-601 4:2:2 resolution. In our experiments, we have also examined the effectiveness of gain and offset correction at these lower bit-rates.

In Tables 8 and 9, 'Train&Tunnel' is a 10 s long sequence which contains the sequences 'Train' and 'Tunnel' (referred to in Tables 1–7) as well as new scene portions. Likewise, the 'Funfair' sequence is a 10 s long sequence, which is a sequence much like an extended version of the 'Manege' sequence.

Since, 'Train&Tunnel' is a relatively easier sequence to encode as compared to 'Funfair', a reasonable SNR (and good visual quality) is obtained in coding left view at 1 Mbit/s. However, for 'Funfair' sequence, the SNR and visual quality are much lower.

From Table 9 it can be seen that gain and offset corrected preprocessing improves coding quality and that the improvements are obviously higher for scenes (and components of the scene) where

Table 9
SNR [dB] for right-view of HHR resolution coded with and without gain-corrected preprocessing with our MPEG-4 proposal (left-view at HHR resolution coded at 1 Mbit/s with $M = 1$ by our MPEG-4 proposal)

Sequence, Bit-rate	Right view disparity- and motion-comp. coded			Gain corrected preprocessed right view disparity- and motion-comp. coded		
	Y	Cb	Cr	Y	Cb	Cr
Train&Tunnel, 512 kbit/s	28.21	33.91	32.94	28.36 (+0.15)	34.20 (+0.30)	34.50 (+1.56)
Funfair, 768 kbit/s	24.80	33.78	32.53	25.16 (+0.36)	33.77 (−0.01)	32.54 (+0.01)

mismatch is higher. In coding the right view, in all cases, only 'Train&Tunnel' is able to achieve the target rate of 512 kbit/s. For 'Funfair' sequence, convergence is only achieved at a higher bit-rate of 768 kbit/s; this is due to the fact that a significant area in 'Funfair' sequence is in motion causing excessive motion vectors that create problems in bit-rate regulation. Due to relatively higher bit-rate used for coding of right view of 'Funfair' as compared to that of 'Train&Tunnel, the SNR achieved for right and left views is quite similar although the SNR of the right view of 'Train&Tunnel' is about 3 dB lower than that for its left view. This means that if it would be possible to code right view of 'Funfair' at 512 kbit/s, its performance would have been similar to that of right view of 'Train&Tunnel', i.e., about 3 dB lower. As mentioned earlier, such differences in quality between the left and the right views are quite tolerable by the human visual system.

In comparing results with and without gain and offset correction in Table 9 we observe that for 'Train&Tunnel' sequence although the difference is rather modest for Y and Cb signals, it is quite significant (> 1.5 dB) for the Cr signal. The difference in 'Funfair' sequence is mainly in luminance signal and is somewhat lower than that found for 'Manege' sequence. Regardless of the SNR differences, the visible improvement with gain and offset correction is quite significant. For 'Funfair' sequence at HHR 4:2:0 resolution, Fig. 14 shows original frames as well as coded frames without and with gain and offset corrected preprocessing, as per our proposal to MPEG-4. Fig. 15 shows similar results for the 'Train&Tunnel' sequence.

9. Multi-viewpoint video

Multi-viewpoint video is simply what is obtained when a scene is imaged from various viewpoints. In fact, stereoscopic video is a special case of multi-viewpoint video with the restriction that the number of views are limited to 2 and that the geometry of imaging these views is guided by the geometry of human binocular vision. Thus, generalized multi-viewpoint video is composed of more than one view, with or without restrictions on how imaging is accomplished. The views of multi-viewpoint video may be arranged such that any adjacent pair of views will form a stereo pair. Multi-viewpoint video can thus provide 'look around' capability similar to holograms.

9.1. Applications of multi-viewpoint video

Since multi-viewpoint video may in fact include stereoscopic video, the applications of multi-viewpoint video also includes applications of stereoscopic video. The new applications of multi-viewpoint video for consumers are still evolving although in some niche applications such as in amusement theme parks and exhibitions, multi-viewpoint video has already been in use for more than 15 years. In any case, we next highlight the important applications of multi-viewpoint video:
- Interactive television,
- Interactive education and training simulators,
- Specialized movies and movie making,
- Medical surgery and surgical planning,
- New generation of video games,

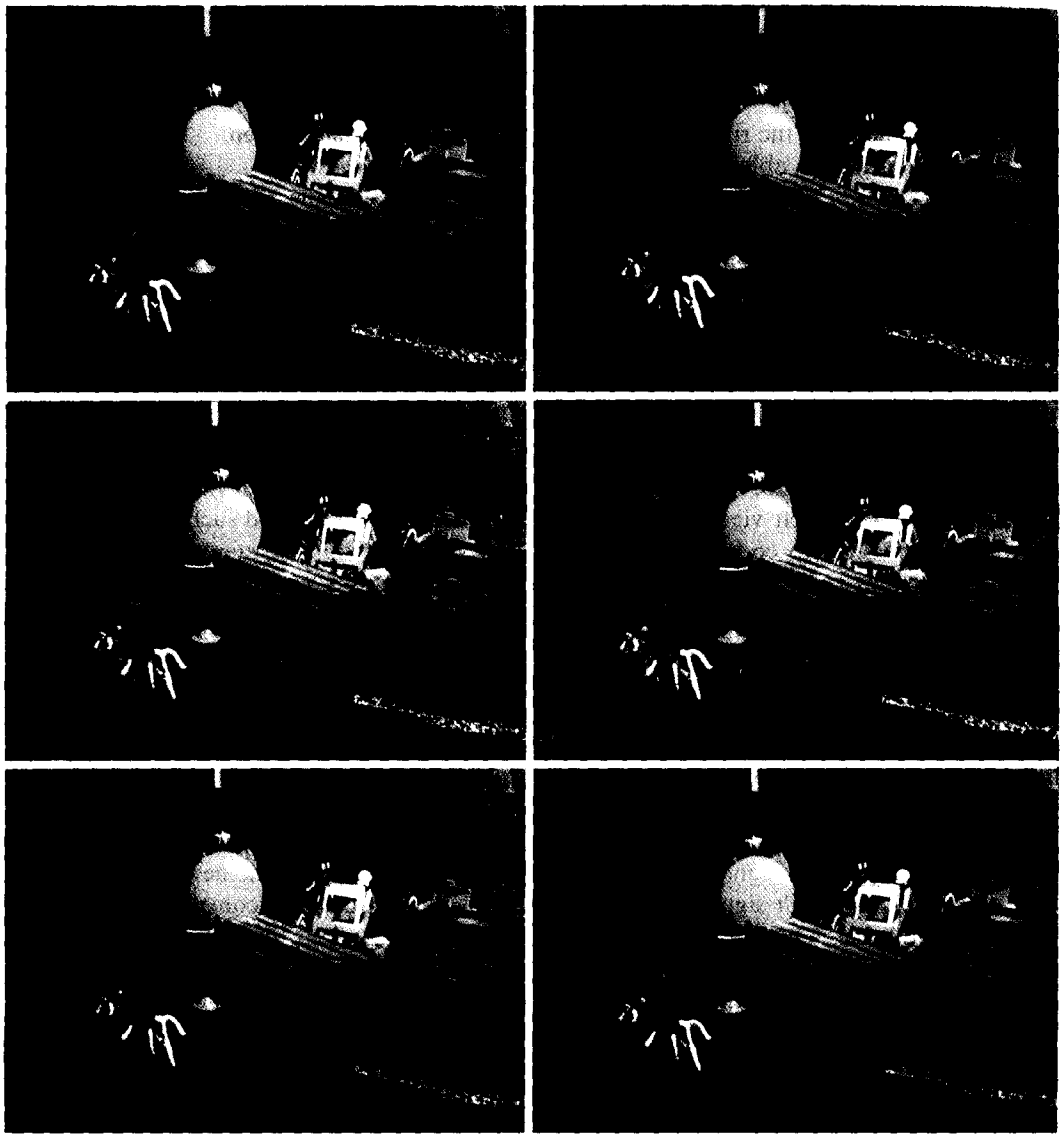

Fig. 12. 'Tunnel' sequence at CCIR-601 4:2:0 resolution: (a) uncoded left view and (b) uncoded right view, (c) left view coded independently with MPEG-2 nonscalable coding at 6 Mbit/s and (d) right view coded with disparity- and motion-compensated coding at 2 Mbit/s, (e) left view coded as in (c) and (f) right view gain and offset corrected by preprocessing prior to coding as in (d).

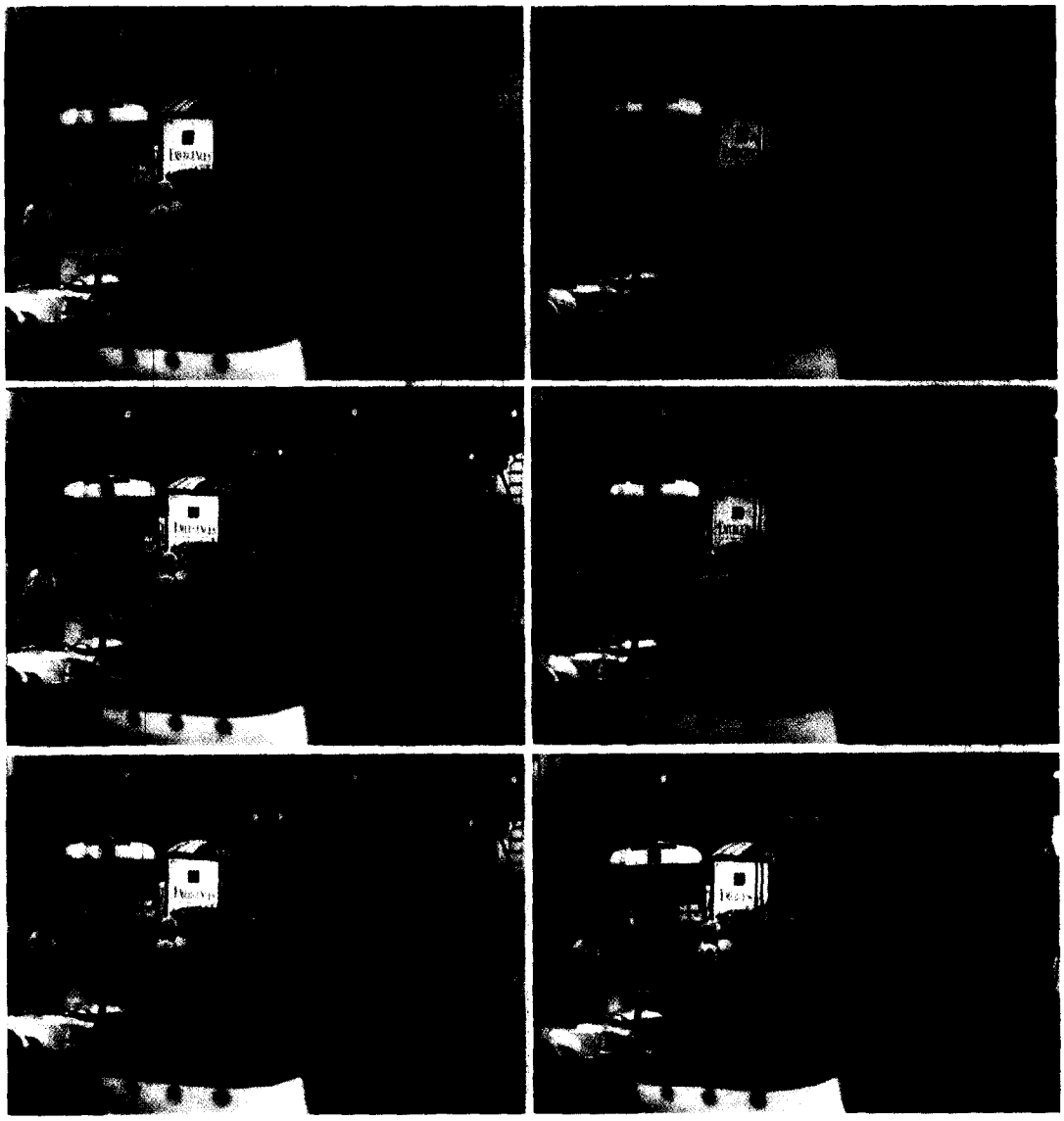

Fig. 13. 'Manege' sequence at CCIR-601 4:2:0 resolution: (a) uncoded left view and (b) uncoded right view, (c) left view coded independently with MPEG-2 nonscalable coding at 6 Mbit/s and (d) right view coded with disparity- and motion-compensated coding at 2 Mbit/s, (e) left view coded as in (c) and (f) right view gain and offset corrected by preprocessing prior to coding as in (d).

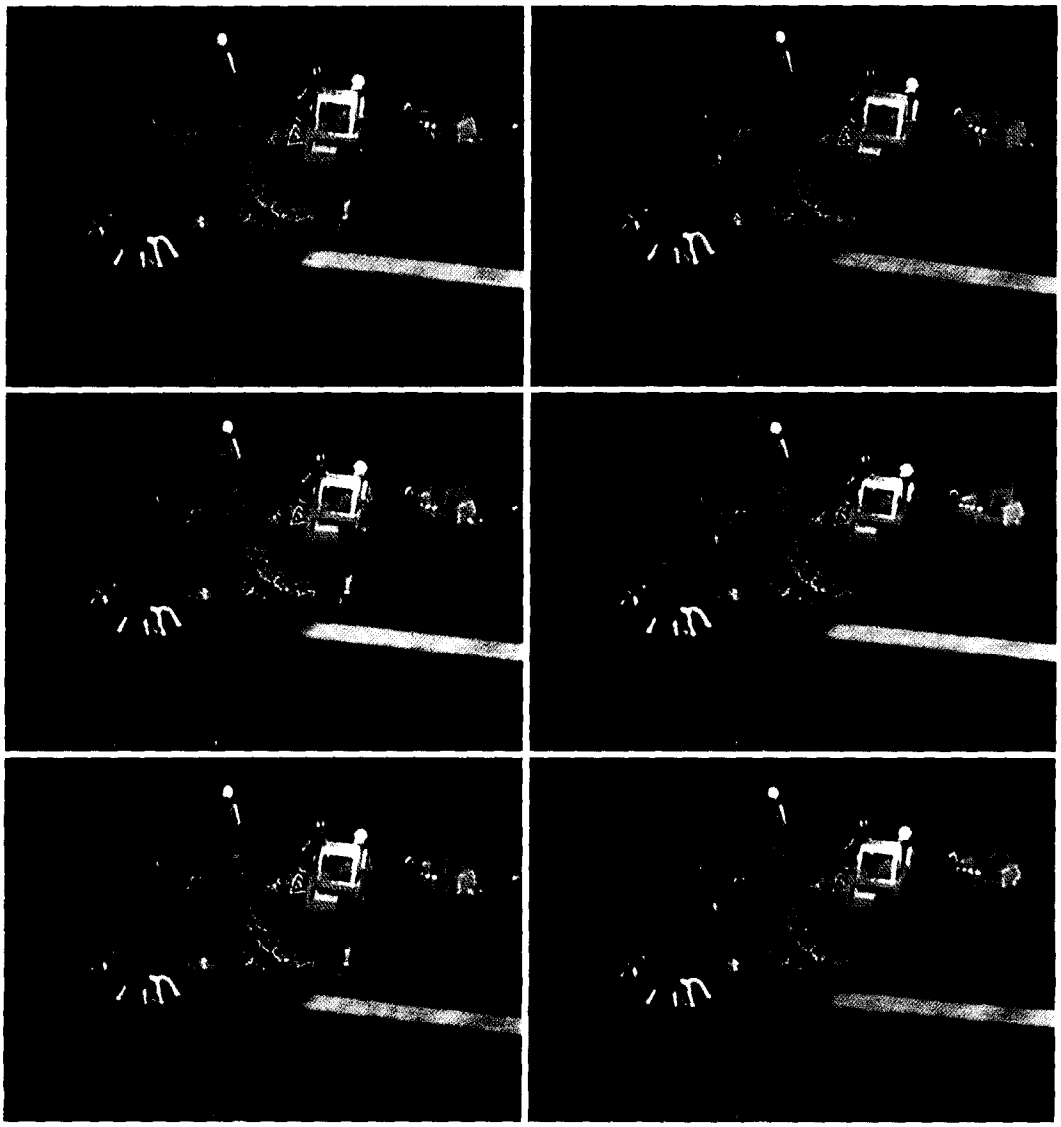

Fig. 14. 'Train&Tunnel' sequence at HHR 4:2:0 resolution: (a) uncoded left view and (b) uncoded right view, (c) left view coded independently with MPEG-2 nonscalable coding at 1 Mbit/s and (d) right view coded with disparity- and motion-compensated coding at 512 Mbit/s, (e) left view coded as in (c) and (f) right view gain and offset corrected by preprocessing prior to coding as in (d).

Fig. 15. 'Funfair' sequence at HHR 4:2:0 resolution: (a) uncoded left view and (b) uncoded right view, (c) left view coded independently with MPEG-2 nonscalable coding at 1 Mbit/s and (d) right view coded with disparity- and motion-compensated coding at 768 Mbit/s, (e) left view coded as in (c) and (f) right view gain and offset corrected by preprocessing prior to coding as in (d).

- Virtual travel and virtual shopping experiences,
- Industrial and remote tele-operations.

The issue of efficient compression of multi-viewpoint video is relatively more complex than that for stereoscopic video due to the increased flexibility in the number and organization of views. The MPEG-2 video committee has agreed to include a solution based on temporal scalability for coding of stereoscopic video as part of the MPEG-2 standard. In general, work is in progress to complete syntax for stereoscopic video coding, which also includes syntax for specifying camera position, orientation, separation, convergence and other parameters [3, 15]. One of the requirements in MPEG-2 is that the syntax be extensible to specify multiple views, although video coding will support only two layers [6]. A simple proposal [15] has been presented to MPEG suggesting how several views can be coded in a two-layer temporal scalability structure. Alternatively, other proposals have also been made [2, 23], however, such schemes are not suitable for MPEG-2 but may be found promising in the future. Currently, progress is being made in MPEG-2 towards supporting stereoscopic video and multi-viewpoint video via the multi-view profile which is discussed next.

9.2. Multi-viewpoint video system

A multi-view video system for transmission and storage is shown in Fig. 16 and derived simply as an extension of the stereoscopic video system of Fig. 5. Unlike the stereoscopic system no constraints need to be imposed on the way the different views of a scene are captured. Of course, in the limiting case of imaging two views of a scene under constraints of stereoscopic viewing, the two cases are equivalent. A multi-view display processor may allow simultaneous display of one or more views of scene depending on the type of display being used and the coded resolution of the scenes. For instance, although the various views of a scene may be imaged at full resolution, to reduce coding complexity and display problems, they could be coded at lower spatial resolution.

9.3. Multi-viewpoint video coding in MPEG-2

In MPEG-2, although many video coding tools are included, they are enabled only by defining a profile. A *profile* is a defined subset of an entire bit-stream syntax of MPEG-2 video. The bit-stream syntax included in a profile corresponds to the coding tools it supports. Further, since a profile may cover a large range of processing complexity, say from the picture resolutions typically used in MPEG-1 to HDTV resolution, profiles are partitioned into levels. Each *level* of a profile represents constraints on the values of parameters in the bit-stream. A profile and level combination explicitly specifies the decoding complexity of a bit-stream. In MPEG-2, 5 hierarchically structured profiles have been defined, each allowing up to 4 levels; however, only 11 combinations are actually allowed. Currently, two new profiles are also being defined that do not have a hierarchical structure with respect to already defined profiles. A profile for multi-viewpoint video is such a profile

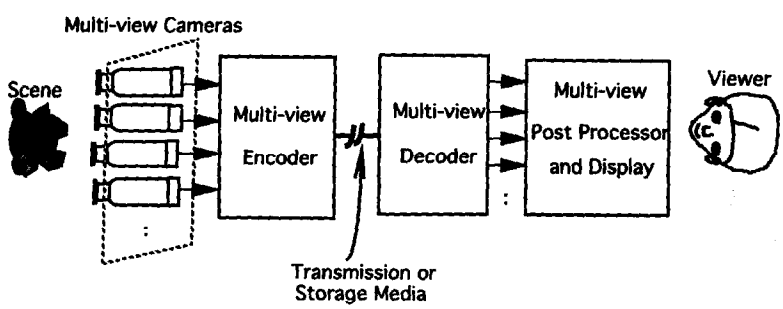

Fig. 16. A multi-view video system.

in progress and is called the multi-view profile (MVP).

The MVP is intended for video applications where more than one view of a scene may need to be coded. It is primarily intended for coding of stereoscopic video but may include flexibilities which make it extensible to more than two views. It will include the temporal scalability video coding tool. It is expected to allow coding of interlaced and progressive video of 4:2:0 format and support conformance at the High, High-1440, Main and Low levels. To keep complexity low, only two layers are being considered, a base-layer and an auxiliary-layer. It has not yet been finalized as to whether each layer can carry more than one view or not. Proposals for the parameters for each level in MVP have been made [14, 15] and as an outcome a Draft Amendment (DAM) [7] to the MPEG-2 video standard has been agreed to. The MVP places a requirement that an MVP compliant decoder only needs to be able to decode two views simultaneously, with one view coded independently and the second view coded with temporal scalability. Although this imposes restrictions on flexibility in dealing with scenes with more than two views to be coded, this restriction was necessary from standpoint of practical decoding hardware. It is expected that the work on MVP will conclude over next several months after extensive bit-stream exchange and verification of coded test sequences and exhaustive subjective testing which is currently in progress. At that stage the MVP in MPEG-2 will advance from DAM to amendment (AM) stage.

9.4. Multi-viewpoint video coding directions in MPEG-4

Currently, work is in progress in MPEG-4 in comparing the most promising components of proposed algorithms via definition of a reference coding scheme, called the verification model (VM), and a set of core experiments. Most of the work up to now has focussed on high efficiency, content-based coding and error resilience functionalities. The stereoscopic and multi-viewpoint coding belong to the multifunctional coding ad hoc group [12] along with several other functionalities such as scalability and object tracking and several experiments are in progress in this group. It is expected that MPEG-4 will address the issue of multi-view coding in more detail allowing efficient compression of more than two (or even four, eight or more) views.

We now envisage an example technique [17] that may be able to handle coding of multi-view video with more than two views. It uses a three-dimensional discrete cosine transform, 3D-DCT. The 3D-DCT can be used to efficiently code B-pictures in the auxiliary layer. To make our discussion more concrete, consider the case of coding of multi-view video consisting of a center-view and 4 surround-views. The center-view can be coded as the compatible signal providing interworking with MPEG-2 and monoscopic displays. The surround views can be predicted using disparity- and motion-compensated prediction with respect to the decoded center view. The resulting prediction errors of temporally coincident frames of each of the 4 surround-views can be used to form $8 \times 8 \times 4$ prediction error blocks which can then be 3D-DCT coded. This concept can be easily extended to deal with more views. Such a scheme can be combined with a scheme for conditionally coding some number of views and interpolating the missing views at the decoder with some assistance information sent via the coded bit-stream.

10. Summary

We have first discussed the various potential applications which could greatly benefit from more realistic imaging and display systems offering the perception of depth in scenes (3D rather than 2D), thus producing an increased sense of presence. As a practical matter, a 3DTV system employing stereoscopic video is believed to be sufficient for many applications. Then, we introduced the basics of 3D/stereo including cues that assist in depth perception, the important issue of techniques for stereoscopic 3D displays, and basic terminology employed in describing the geometry of stereoscopic imaging and display.

This was followed by an introduction to basics of MPEG-2 video coding with special emphasis on tools relevant to coding of stereoscopic video such

as the temporal scalability tool. The various potential approaches for coding of stereoscopic video such as simulcast, compatible and joint coding were discussed next. This was followed by a detailed discussion of a compatible stereoscopic coding approach based on temporal scalability including various forms of predictions such as disparity-based prediction and disparity- and motion-based prediction. Next, a technique based on preprocessing of stereoscopic video for improving the compression efficiency and display quality was introduced. Considerations for low bit-rate coding of stereoscopic video for MPEG-4 applications were then introduced. A coding scheme was derived for MPEG-4 which uses modified MPEG-2 coding, mismatch-corrected preprocessing, scene and bandwidth adaptive preprocessing and temporal rate adaptive postprocessing.

Further, it was demonstrated via experiments on stereoscopic video scenes that the temporal scalability of the MPEG-2 video standard can indeed be successfully applied for efficient compression of stereoscopic video. Combined disparity- and motion-compensated prediction significantly outperforms other techniques against which comparisons have been made. Further improvements can be obtained by gain-corrected preprocessing. Overall, stereoscopic video of full resolution with good quality can be obtained in a total of 6 to 8 Mbit/s. Coding results are also included at low bit-rates for the scheme proposed for MPEG-4.

Then, we discussed generalization of stereoscopic video to multi-viewpoint video and its various applications. The ongoing activity in MPEG-2 towards defining a profile for stereoscopic and multi-view video was summarized. Finally, we introduced the relevance of developing MPEG-4 standard to coding of multi-viewpoint video and suggested some directions that this effort is likely to take.

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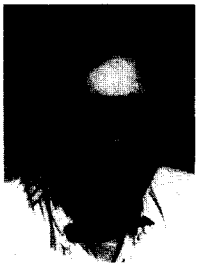

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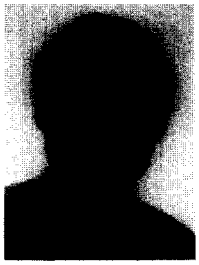

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